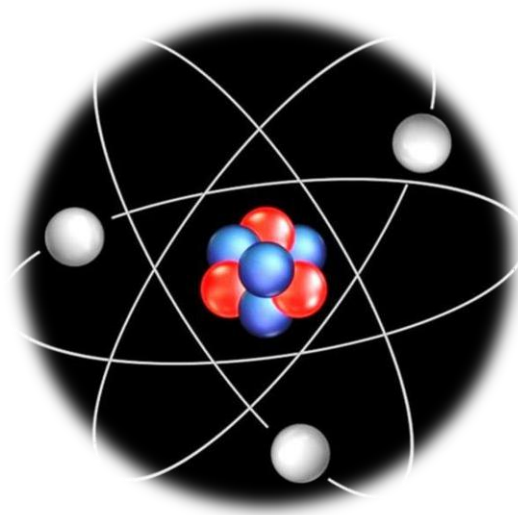


**Department of Science and Humanities
Applied Science Laboratory**



Laboratory Manual

(As per Revised Curriculum SVU R-2025)

FY B Tech AY 2026-27 SEMESTER I

List of Experiments:

Title of the Experiment
Grating Constant
Newton's ring
Numerical Aperture
Energy Band Gap
Planck's Constant

Note:

In order to successfully complete the Lab CA in Physics Lab Course (Course code: 316U06L102), students need to perform and prepare a written record of experiments.

Instructions:

1. **Take printouts of only the template (page no 14 & 15) for each experiment.** The template for preparing a written record of experiments is given at the end of this manual. You can keep a soft copy of the lab manual for reference during the lab sessions on your laptop/mobile phone.
2. Observation tables, graphs, calculations etc. should be handwritten. A neatly written and well-organized journal will fetch better grades.
3. Results and/or conclusions should be written for every experiment.
4. Attach graph paper separately. Axes should be labelled. Indicate scale used and units.
5. Attach a separate sheet for assignment work.
6. Each experiment should be completed and submitted on time as instructed by the Faculty I/c. Grades will be deducted for late submissions.
7. **Lab attendance should be 100%. Grade penalties will be applied for absenteeism.**
8. Journal carries 30 marks out of 50* marks of Lab CA. Distribution of marks for each experiment will be as follows:

Component	Max marks	Remarks
Performance, written record of experiment and its verification	20	Draw graph on a graph paper and attach with written record of the experiment
Assignment	05	Write assignment Q&A on separate assignment sheet
Attendance	05	0 for absent, 03 if performed in repetition turn
Total	30	Average of all experiments will be considered for grading the journal

***remaining 20 marks are allotted to Lab quizzes based on experiments performed**

Title of the experiment: Grating Constant

Aim: To determine the Wavelength (λ) of He-Ne laser light.

Apparatus: He-Ne Laser apparatus, Grating, Scale.

Link : <https://ep-iitb.vlabs.ac.in/exp/diffraction-grating/theory.html>

Diagram:

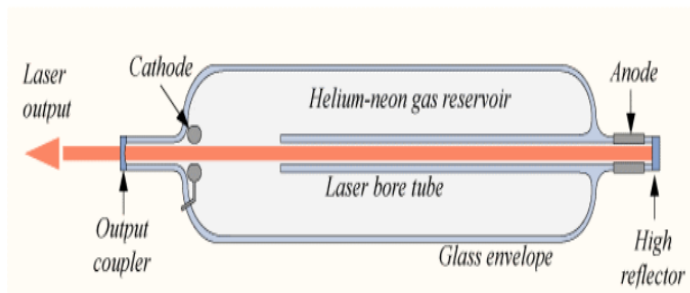


Fig.1. He-Ne Laser

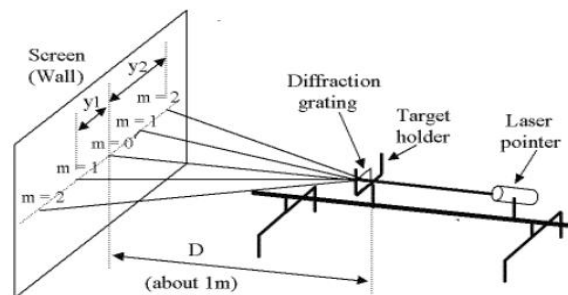


Fig.2. Diffraction Pattern

Procedure:

1. Switch on the electric power supply which is given to He-Ne Laser Apparatus.
2. Observe the diffraction pattern on the Screen.
3. Measure the distance between grating & the Screen (D).
4. For the 1st order spectrum, measure the distance y_1 & y_2 .
5. Similarly, the 2nd order spectrum also measures y_1 & y_2 .
6. From the available data, compute wavelength (λ) for the given He-Ne Laser light.

Observations:

Table 1:

Screen distance D = _____ cm Grating type: _____								
Sr. No.	Order of spectrum (n)	Distance between Grating & screen in cm (D)	Left (y_1) cm	Right (y_2) cm	Mean Y = $(y_1 + y_2)/2$	$\theta = \tan^{-1} \frac{Y}{D}$	Sin θ	$\lambda = \{(a+b) \sin \theta\}/n$
1								
2								
3								

Table 2:

Screen distance D = _____ cm Grating type: _____								
Sr. No.	Order of spectrum (n)	Distance between Grating & screen in cm (D)	Left (y ₁) cm	Right (y ₂) cm	Mean Y = (y ₁ + y ₂)/2	$\theta = \tan^{-1} \frac{Y}{D}$	Sin θ	$\lambda = \{(a+b) \sin \theta\} / n$
1								
2								
3								

Formula:

$$n\lambda = (a + b) \sin \theta$$

Where,

n = order of wavelength

λ = wavelength of laser beam

θ : angle of diffraction maximum,

$$\theta = \tan^{-1} \left(\frac{Y}{D} \right)$$

(a + b) = grating element

$$(a + b) = 2.54 / 15000 = 1.69 \times 10^{-4}$$

Graph: Plot sin θ as a function of order of diffraction maxima (n). From the points plotted, draw the best fit line. Determine slope of this line.

Calculation: Determine the wavelength line density of the diffraction grating used in the experiment using the formula given.

Result/s and Conclusion:

1. Wavelength (λ) of a given light is _____ Å
2. Standard value of wavelength (λ) of a given light is _____ Å
3. Percent Error in obtained result _____ %

Assignment:

1. Explain the principle of diffraction of light using a plane diffraction grating, and define the grating element (a+b).
2. He-Ne laser beam of wavelength = 632.8 nm is incident normally on a diffraction grating with a grating element (a+b) = 1.693×10^{-6} . Calculate the angle of diffraction for the 2nd order diffraction maximum n = 2.

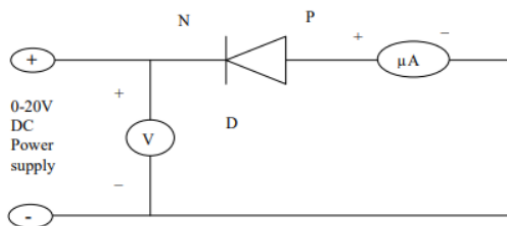
Title of the experiment: Energy Band Gap

Aim: To determine energy band gap of semiconductor p-n junction diode using temperature dependence of reverse saturation current

Apparatus: DC power supply, voltmeter, micro-ammeter, semiconductor diode, heating bath, thermometer, and connecting wires.

Link : <https://bop-iitk.vlabs.ac.in/exp/energy-band-gap/index.html>

Diagram



Procedure:

- 1) Click the “Enable Power” button.
- 2) Choose the correct image.
- 3) After the correct answer “X” (cross) the image.
- 4) Click the “Place Container” button and then click on the drawing box.
- 5) Click on the “Battery” button and then click on the drawing box.
- 6) Click on the “Ammeter” button and then click on the drawing box.
- 7) Click on the “Voltmeter” button and then click on the drawing box.
- 8) Click on the “PN Junction” button and then click on liquid inside the container.
- 9) Click on the “Point” button and then click on the drawing board wherever point connections are required.
- 10) Click on the “Connect” button and then click on the point to be connected to complete the circuit diagram.
- 11) Click on the “Undo/Redo” button in case of wrong connections.
- 12) Select voltage from the drop down button.
- 13) After selecting voltage press the “Click” button.
- 14) Draw the slider to choose the temperature.
- 15) Click on the “Find here” button to get reverse saturation current.
- 16) Repeat this process to complete the observation table.
- 17) Click the “Calculate Data” button.
- 18) Click on the “Draw Graph” button.
- 19) Place the cursor on the plotted graph to enter the value of the first and second coordinate.
- 20) Click on the “Find Slope” button.
- 21) Enter the value of the calculated slope.
- 22) Click “Calculate” button to get the value of Energy Band Gap

Observations:

Sr. No	Temperature (T)		Reverse current I_s (amp)	1000/T (K ⁻¹)	Log ₁₀ I_s
	°C	kelvin			
1	RT				
2	40				
--	--				
--	--				
--	--				
8	70				

Formula:

Energy Band Gap $E_g = |\text{slope}| / 5.04$

| Slope | : absolute value of slope of line $\text{Log}_{10} I_s$ v/s $\left(\frac{1000}{T}\right)$

k: Boltzmann constant. $\kappa = 1.38 \times 10^{-23}$ J/K

I_s : reverse saturation current

T: absolute temperature

$$I_s = A \cdot T^{3/2} e^{-\frac{E_g}{KT}}$$

$$\log_{10} I_s = \text{Constant} - 5.04 \cdot E_g \cdot 10^3 \cdot \frac{1}{T}$$

$$E_g = \frac{\text{Slope of the line}}{5.04}$$

Graph: Plot $\text{Log}_{10} I_s$ as a function of $\left(\frac{1000}{T}\right)$. The graph will be in the 2nd quadrant. From the points plotted, draw the best fit line. Determine slope of this line. Use the absolute value of the slope for calculations.

Calculation: Determine energy band gap of the material of given p-n junction diode using the formula given.

Result/s and Conclusion:

Energy band gap diode (E_g) = _____ eV.

Assignment:

1. Draw a neat and labelled diagram of the experiment.
2. Why paraffin oil bath is used for heating not water bath during the experiment.
3. In the above experiment, explain the process if the diode is kept under forward bias condition

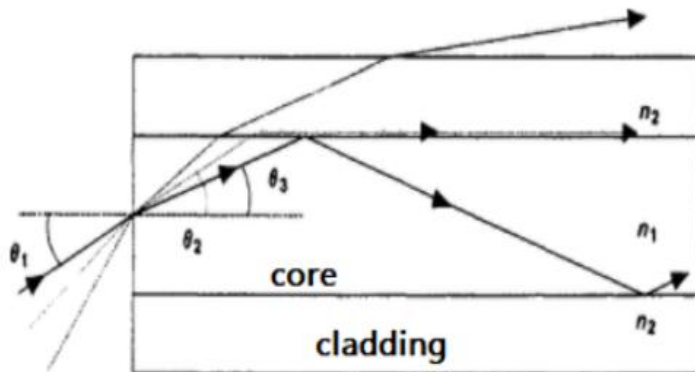
Title of the experiment: Numerical Aperture

Aim: To Measure Numerical aperture and attenuation constant of optical fibre

Apparatus: Optical fibre cable, Fibre stand, Emitter (LED output), output screen.

Link: <https://bop-iitk.vlabs.ac.in/exp/numerical-aperture-measurement/index.html>

Diagram:



Procedure:

1. Click on the “Emitter” button.
2. Click on “Fiber Stand”
3. Click on “Fiber Cable”
4. Click on “Output Screen”
5. Click the “Start Button”
6. Select Fiber cable/Glass cable
7. Drag the slider to set distance of screen in mm. Note the diameter on the “Message” screen
8. Enter the value distance of screen and diameter in the table.
9. Repeat the procedure to complete the data table
10. Click on “Draw Graph” button. Enter the standard value of numerical aperture in the “Final Numeric Aperture” box.
11. Click on the “verify” button to find the percentage error.

Observation Table:

Observations:			
Sr. No.	Distance of screen L (mm)	Dimensions of the illuminated spot (mm)	N.A
		D	
1			
2			
3			
4			
5			
6			
Average NA =			

Formulae:

$$\text{N.A.} = \sin [\tan^{-1} (D/2L)]$$

Where:

NA: Numerical Aperture

L : Distance of fibre tip from screen

D : Diameter of the spot

Results & Conclusion:

1) NA of given optical fibre is _____

Assignment:

1. Explain how the refractive indices of the core and cladding in an optical fiber affect its numerical aperture and acceptance angle.
2. In an optical fibre the refractive indices of core and cladding are 1.46 and 1.44 respectively. Find the angle of acceptance of the optical fibre. If the entire set up is kept in water of refractive index 1.33, discuss what will happen to numerical aperture and acceptance angle.

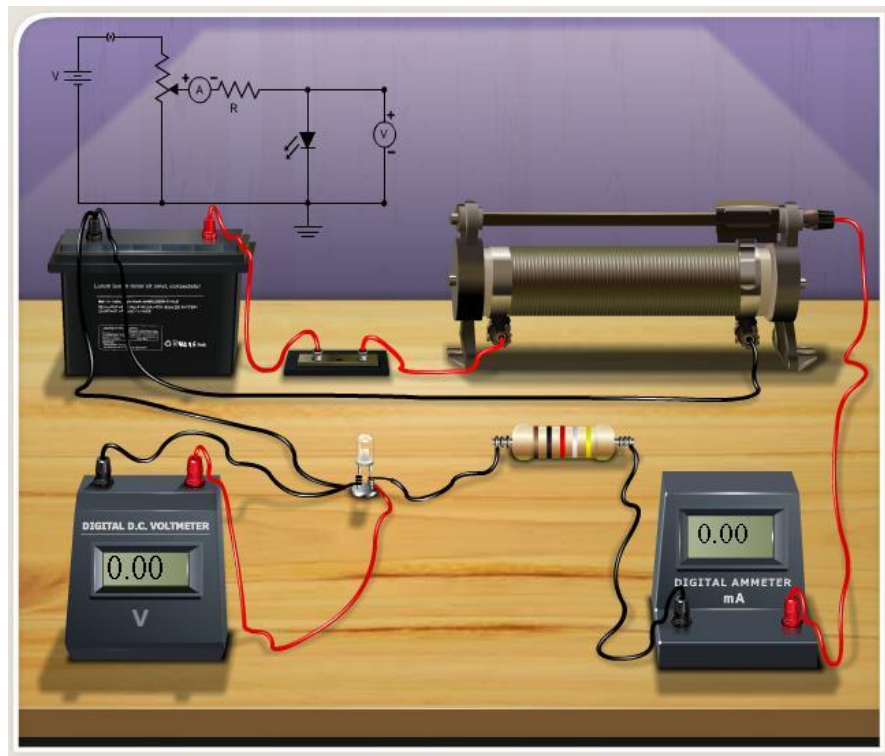
Title of the experiment: Planck's constant

Aim: To determine the value of Planck's constant using LEDs of different colors

Apparatus: Power supply, rheostat, milliammeter, voltmeter, 1 K resistor, various LEDs.

Link: <https://mp-amrt.vlabs.ac.in/exp/determination-plancks-constant/>

Diagram:



Procedure:

- 1) Connect the circuit as shown in the diagram. After the connections are completed, click on 'Insert Key' button.
- 2) Click on the combo box under the 'Select LED' button.
- 3) Click on the 'Rheostat Value' to adjust the value of rheostat.
- 4) Corresponding voltage across the LED is measured using a voltmeter, which is the knee voltage.
- 5) Repeat, by changing the LED and note down the corresponding knee voltage.

Observations:

Sr. No.	LED	Wavelength λ (nm)	$1/\lambda$ (nm ⁻¹)	V_{th} (volt)
1.	RED			
2.	YELLOW			
3.	GREEN			
4.	BLUE			
5.	IR			

Formula: $h = \text{slope} \times (e/c)$

Slope: slope of line V_{th} v/s $1/\lambda$

λ : wavelength of light for a particular

LED: charge ' e ' = 1.6×10^{-19} C

$c = 3 \times 10^8$ m/s

Graph: Plot a graph of threshold voltage (V_{th}) as a function reciprocal of wavelength ($1/\lambda$). Take V_{th} along the Y-axis and $1/\lambda$ along the X-axis. From the points plotted, draw best-fit line. Determine slope of this line. Also mark points corresponding to threshold voltage for IR and Blue LEDs.

Calculation: Determine Planck's constant using the formula given. Find wavelengths of IR and blue LEDs from the marked points.

Result:

Value of Planck's constant (h) = _____ Js

Assignment:

Measure the threshold voltage for IR and Blue LEDs. Mark points corresponding to threshold voltage on the graph plotted. Find wavelengths of IR and blue LEDs from the marked points.

Wavelength of IR LED = _____ nm

Wavelength of Blue LED = _____ nm

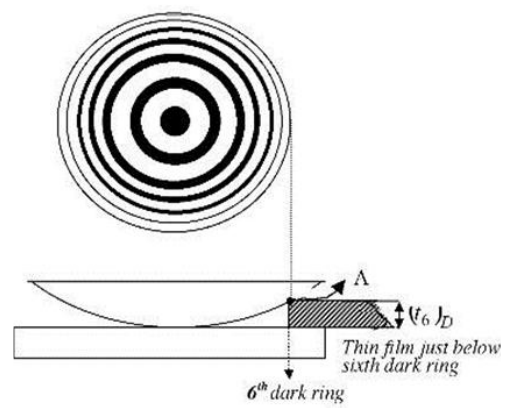
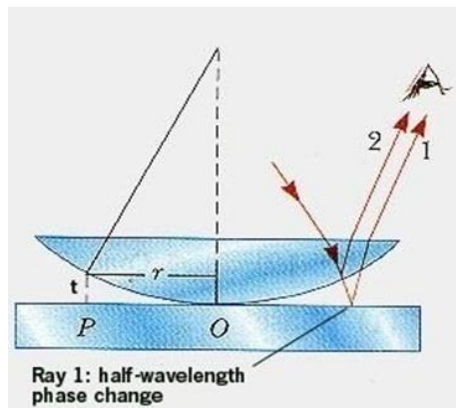
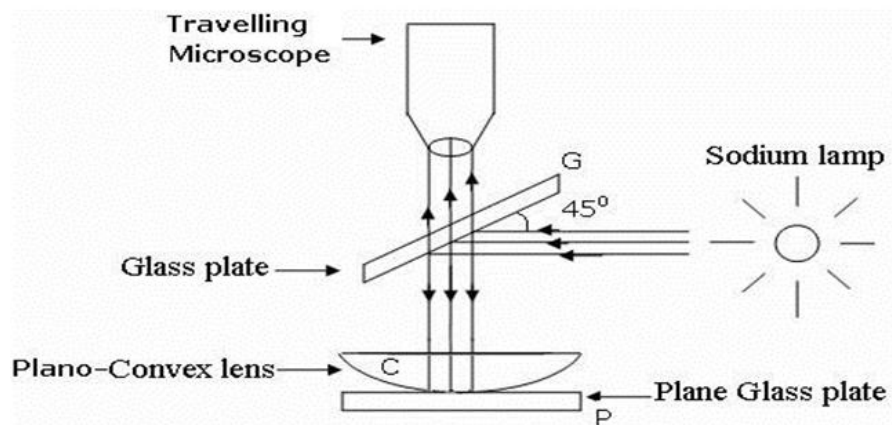
Title of the experiment: Newton's Rings

Aim: To determine wavelength (λ) of light by Newton's rings Experiment.

Apparatus: Travelling microscope, sodium vapor lamp, Newton's ring apparatus

Link : <https://ep-iitb.vlabs.ac.in/exp/newtons-ring-experiment/index.html>

Diagram:



Precaution:

1. Glass plate & lens should be clean to avoid the presence of dust particles between them.
2. The lens should be of a large radius of curvature.
3. The glass plate should be inclined at an angle of 45° with an incident light.
4. Cross wire should be focused tangentially on the ring.
5. The source should be an external source.

Procedure:

1. Find the least count of travelling microscope.
2. Arrange the apparatus as shown in fig so as to obtain Newton's ring.
3. Select the ring; say 1st and 2nd for determining their radius.
4. Set the vertical cross wire tangential to the 1st ring (left side first) Note down the readings.
5. Repeat the step (4) for the 2nd ring (left side).
6. Set the cross wire on the 1st ring (Now on the ring side) & again note down the microscopic reading, repeat the same procedure for the 2nd ring (right side).
7. Calculate the wavelength of light used.

Where,

M - Main Scale Reading.

V - Vernier Scale Reading.

TR - Total Reading.

Observations:

Sr. No	Ring no (n)	Travelling Microscope Reading (cm)						Diameter D _n (cm)= L-R	D ² _n
		*On Left L = M + V (cm)			*On Right R = M + V (cm)				
		MSR	VSR	L	MSR	VSR	R		
1	6								
--	5								
--	4								
—	3								
—	2								
7	1								

*M: Main scale reading, V: Vernier reading = coinciding division $\times$ least count, L: Total reading on left side and R: Total reading on right side of the centre of the ring pattern

Formula:

Slope: Plot a graph D_n^2 v/s n and find the slope of the graph.

$$\lambda = \frac{D_m^2 - D_n^2}{4(m - n)R}$$

D_m = Diameter of m^{th} ring.

D_n = Diameter of n^{th} ring.

λ = Wavelength of light.

R = Radius of curvature of lens

Graph: Plot a graph of squares of diameters of dark rings as a function of dark ring number. Take D^2 along the Y-axis and n along the X-axis. From the points plotted, draw the best-fit line that passes through the origin also. Determine slope of this line.

Calculation: Determine the wavelength of the light used as per given formula.

Result/s and Conclusion:

1. Wavelength (λ) of a given light is _____ Å
2. Standard value of wavelength (λ) of a given light is _____ Å
3. Percent Error in obtained result _____ %

Assignment:

1. Why is the central spot of the Newton's rings interference pattern dark when viewed in reflected light?
2. In a Newton's rings experiment, the measured diameters of the 6th and 2nd dark rings are $D_6 = 0.38\text{cm}$ and $D_2 = 0.22\text{ cm}$ respectively. Calculate the wavelength of the light source if the radius of curvature of the lens is $R = 100\text{ cm}$.

Name:
Roll Number:
Batch:
Experiment performed on (date):

Title of the Experiment:

Aim:

Apparatus:

Observations:

Calculations:

Result/s and Conclusion/s: